Proiectare Bandgap Brokaw

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1. Proiectarea structurii Bandgap

1.1. Specificații

Caracteristică	Valoare
Tip Bandgap	Brokaw
Tensiune Bandgap	1.2V
Buget de curnet	20µA
Domeniu temperatură	-40:150degC
Precizie în temperatură	±3%
Domeniu alimentare	1.6:2.0V
Tip OTA	Miller

Tabel 1. Specificații generale

1.2. Dimensionare OTA Miller

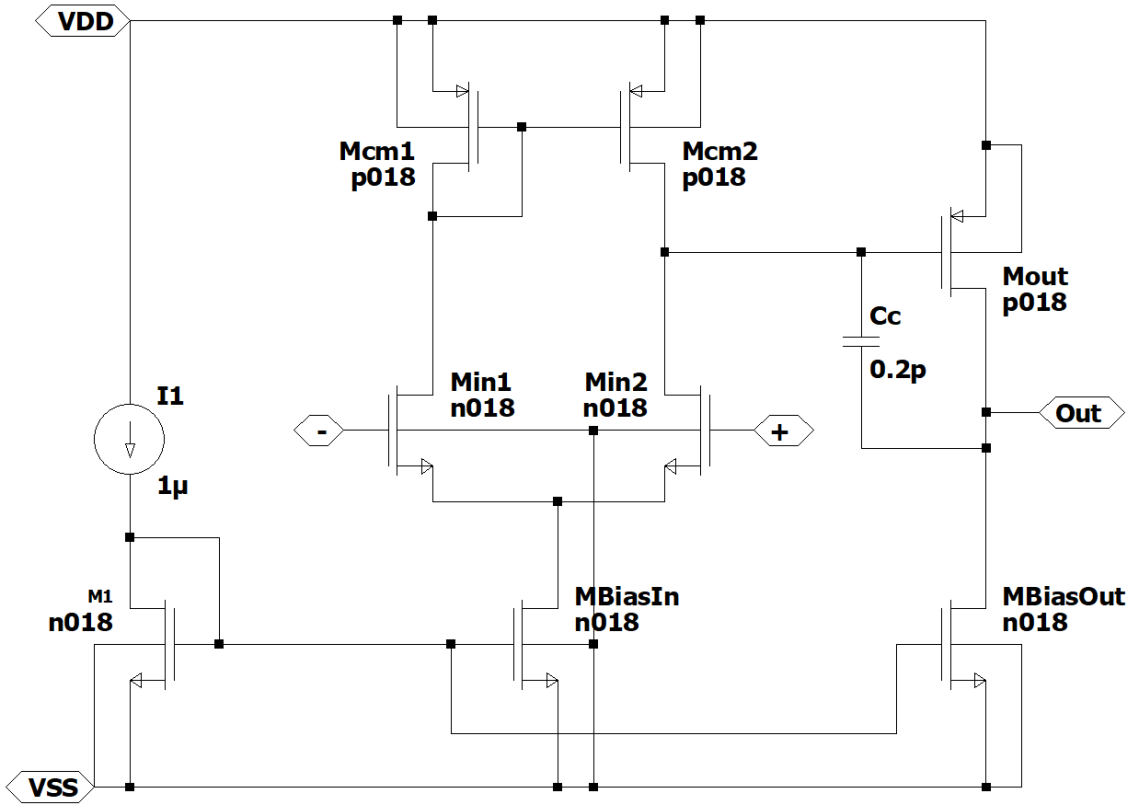


Figura 1. Circuit OTA Miller

$$C_C = 0.2C_L = 0.2 \times 1pF = 0.2pF$$

Transconductanța de intrare (GBW=15MHz)

$$g_{mIN} = 2\pi C_C GBW = 2\pi \times 0.2pF \times 15MHz = 18.8\mu S$$

Transconductanța etajului de ieșire

$$g_{mOUT} = 1.5 \times GBW \times 2\pi C_L =$$
$$= 1.5 \times 15MHz \times 2\pi \times 1pF = 141\mu S$$

1.2.1. Determinarea curenților

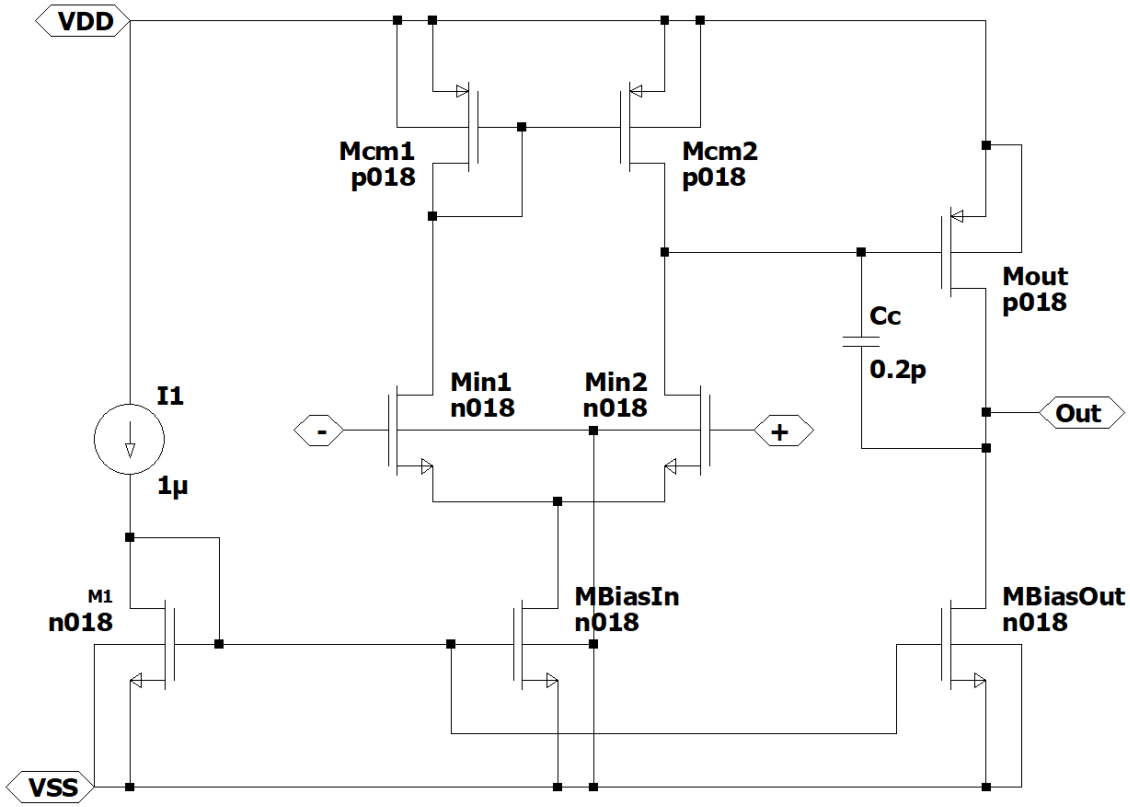


Figura 1. Circuit OTA Miller

$V_{DsatIN} = 100mV\ (ales) =>$
$i_{BiasIN} = V_{DsatIN} \times\ g_{mIN} = 1.88\mu A$

$V_{DsatOUT} = 100mV(ales) =>$
$i_{BiasOUT} = 0.5 \times V_{DsatOUT} \times\ g_{mOUT} = 7.05\mu A$

Verificare buget de curent:

$i_{BiasOUT} + i_{BiasIN} \leq 10\mu A \Rightarrow\ 7.05\mu A +\ 1.88\mu A\ \leq 10\mu A$

1.2.2. Set de referință

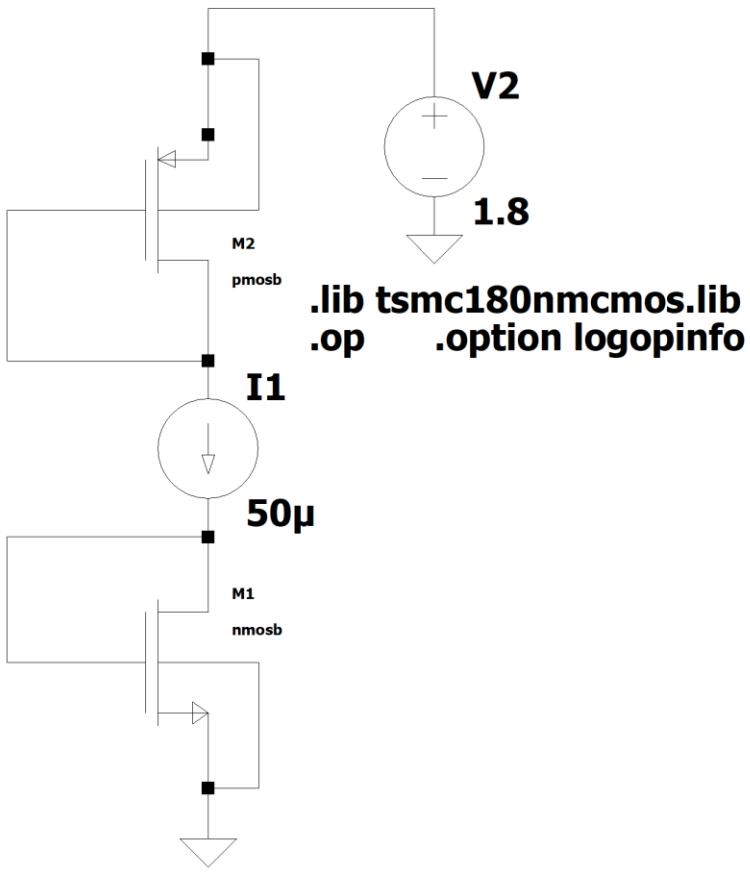


Figura 2. Circuit set referință

Tranzistor	I_D [µA]	V_{Dsat}[mV]	W	L
NMOS	50	145	1.8µm	180nm
PMOS	50	156	7.2µm	180nm

Tabel 2. Set de referință pentru dimensionarea tranzistoarelor

1.2.3. Dimensionare tranzistoare

$$V_{Dsat} = 100mV \; si \; L = 450nm$$

$$\frac{W}{L} = \frac{W_{Ref}}{L_{Ref}} \times \frac{i_D}{i_{DRef}} \times \left(\frac{V_{DsatRef}}{V_{Dsat}}\right)^2$$

$$\left(\frac{W}{L}\right)_n = \frac{1.8\mu m}{180\text{nm}} \times \frac{1\mu m}{50\mu m} \times \left(\frac{145}{100}\right)^2 = \frac{840n}{2\mu}$$

$$\left(\frac{W}{L}\right)_p = \frac{7.2\mu m}{180\text{nm}} \times \frac{1\mu m}{50\mu m} \times \left(\frac{156}{100}\right)^2 = \frac{3.9\mu}{2\mu}$$

Tranzistor	M_1	M_{BiasIN}	M_{IN1}	M_{IN2}	$M_{CM1}[p]$	$M_{CM2}[p]$	$M_{out}[p]$	$M_{BiasOut}$
Multiplicator	1	1.88	0.94	0.94	0.94	0.94	7.05	7.05
W	840n	1.59µ	794n	794n	3.66µ	3.66µ	27.45µ	5.96µ
L	2µ	2µ	2µ	2µ	2µ	2µ	2µ	2µ

Tabel 3. Dimensiunile tranzistoarelor interne

1.3. Dimensionare Bandgap

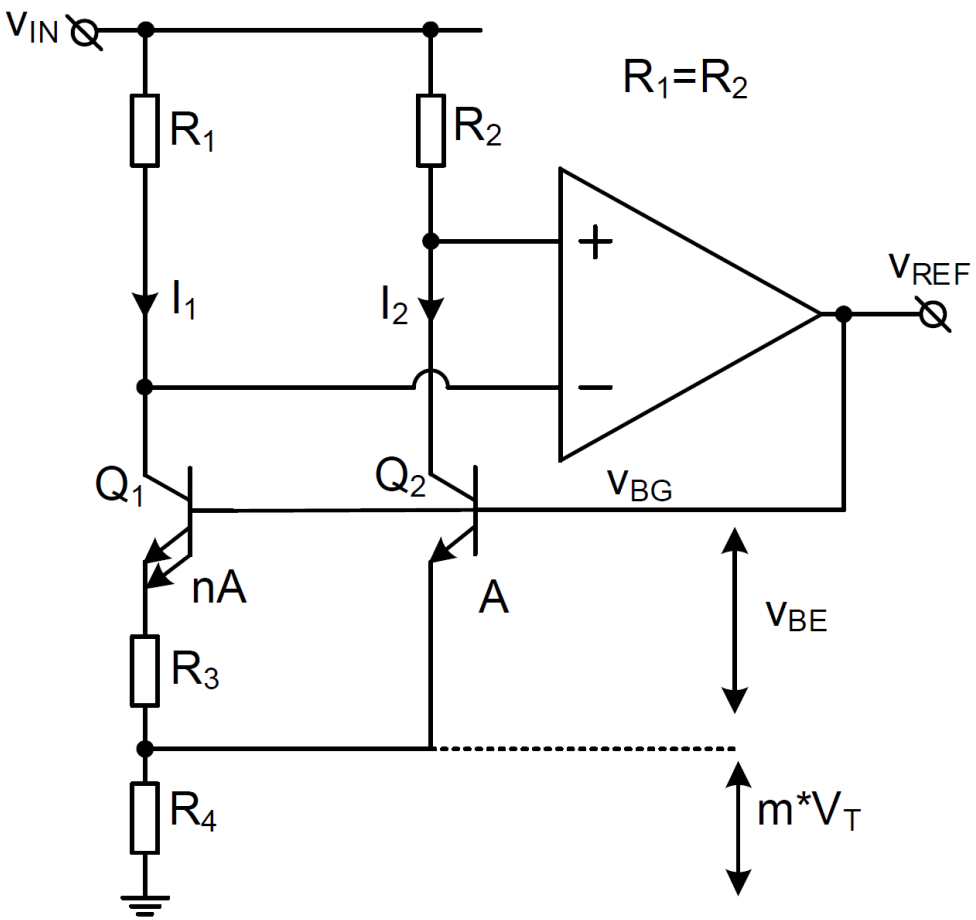


Figura 3. Circuit referință tip Brokaw

$$v_{OA}^{+} = v_{OA}^{-} \rightarrow v_{R1} = v_{R2} \xrightarrow{R1=R2} I_{C1} = I_{C2}$$

$$v_{R_3} = v_{BE2} - v_{BE1} = V_T \ln n$$

$$I_{R3} = \frac{v_{R_3}}{R_3} = \frac{V_T}{R_3} \ln n \; ; \; I_{R3} = I_{E1}$$

$$I_{E1} \cong I_{C1} \; ; \; I_{E2} \cong I_{C2} \; (\beta >> 1)$$

$$V_{REF} = V_{BG} = v_{BE2} + v_{R_4} = v_{BE2} + R_4 \cdot (I_{C1} + I_{C2})$$

$$V_{BG} = \underbrace{\boxed{v_{BE2}}}_{\text{CTAT}} + \underbrace{\boxed{V_T \cdot \frac{2R_4}{R_3} \ln n}}_{\text{PTAT}}$$

$$\boxed{\frac{\partial V_{BG}}{\partial \theta} = 0 \Rightarrow \frac{\partial V_{BE2}}{\partial \theta} = -\frac{k}{q}\frac{2R_4}{R_3} \ln n}$$

Condiția de deriva termică nulă

1.3.1. Dimensionare rezistori și raport Q_1/Q_2

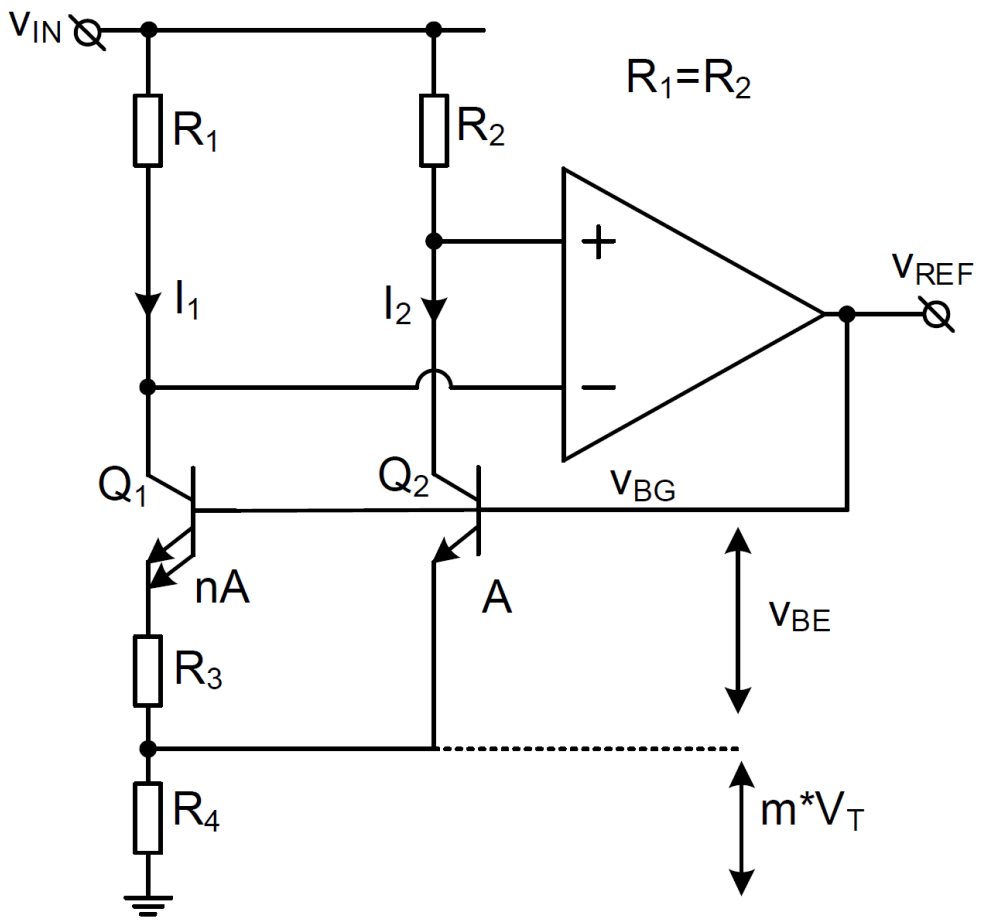


Figura 3. Circuit referință tip Brokaw

$$\left.\begin{aligned}
&\frac{\partial V_{BG}}{\partial \theta} = 0 \Rightarrow \frac{\partial V_{BE\,2}}{\partial \theta} = -\frac{k}{q}\frac{2R_4}{R_3}\ln n \\
&\frac{\partial V_{BE\,2}}{\partial \theta} = -2\frac{mV}{1K} \\
&\frac{\partial V_T}{\partial \theta} = \frac{k}{q} = +0.087\frac{mV}{1K}
\end{aligned}\right\} \Rightarrow -2 = -0.087\frac{2R_4}{R_3}\ln n \Rightarrow$$

$$23 = \frac{2R_4}{R_3}\ln n \xrightarrow[n=10]{} \frac{R_4}{R_3} = 5 \Rightarrow \begin{cases} R_4 = 591.5K\Omega \\ R_3 = 120K\Omega \end{cases}$$

$$\left.\begin{aligned} I_1 = I_2 \\ \text{Buget de curent} = 10\mu A \end{aligned}\right\} \xrightarrow[v_{in}=2V]{} R_1 = R_2 = 400K\Omega$$

1.4. Dimensionare circuit Start-up

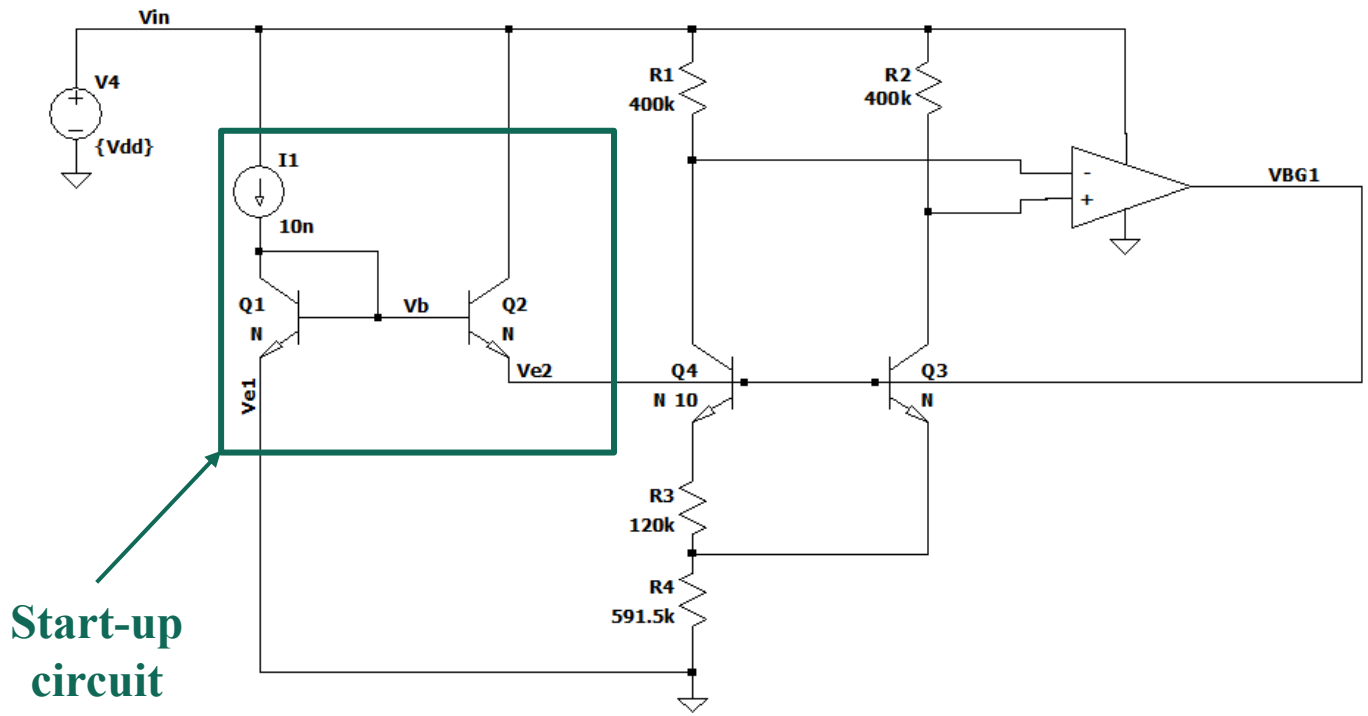


Figura 4. Circuit Start-up

Principiu de funcționare

$V_{BE_1} = V_{CE_1}$ (configurație diodă)

$$V_{BE_1} = V_{CE_1} = V_T \ln \frac{I_C}{I_S} \xRightarrow[\substack{*I_S=0.1\,fA \\ I_C=10nA}]{} V_{CE_1} = 478mV$$

$V_{CE_1} = 478mV > 200mV \Rightarrow Q_1$ în a_F

$V_{B_2} = V_{CE_1}$; $V_{E_2} = V_{BG} = 0\ (inițial)$ $V_{C_2} = V_{in}$

$V_{CE_2} > 200mV \Rightarrow Q_2$ în a_F

Atât timp cât tranzistoarele se află în regiunea activă oglinda injecteză curent în circuit.

Când V_{BG} crește $\Rightarrow V_{BE_2}$ scade $\Rightarrow$
$\Rightarrow Q_2$ iese din a_F și se blochează

** LTspice uses default I_S =1.0E-16*

1.4.1. Circuit Start-up principiu funcționare

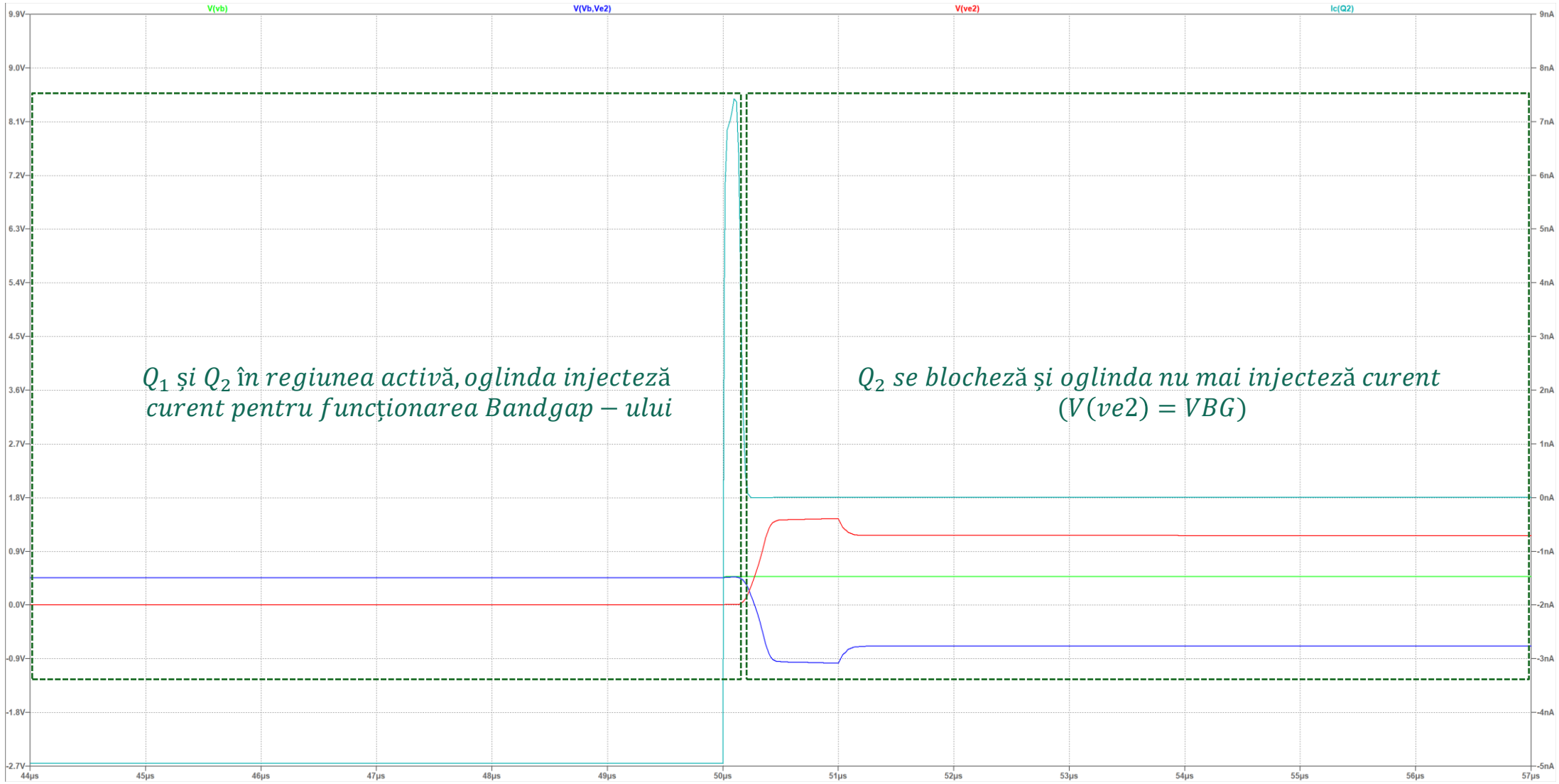


Figura 5. Funcționare circuit Start-up

2. Caracteristici și rezultate de simulare

2.1. V_{BG} vs. temperatură

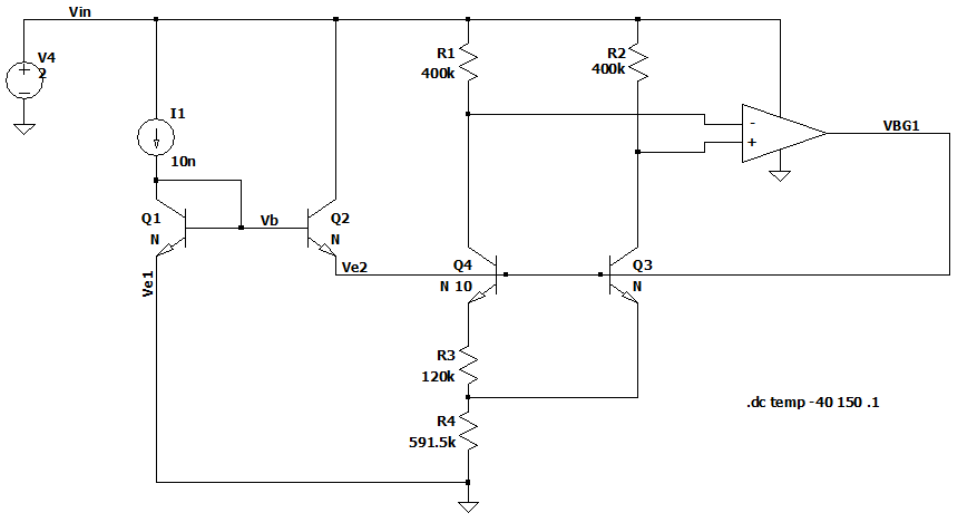


Figura 6. Circuit Bandgap + analiză de temperatură

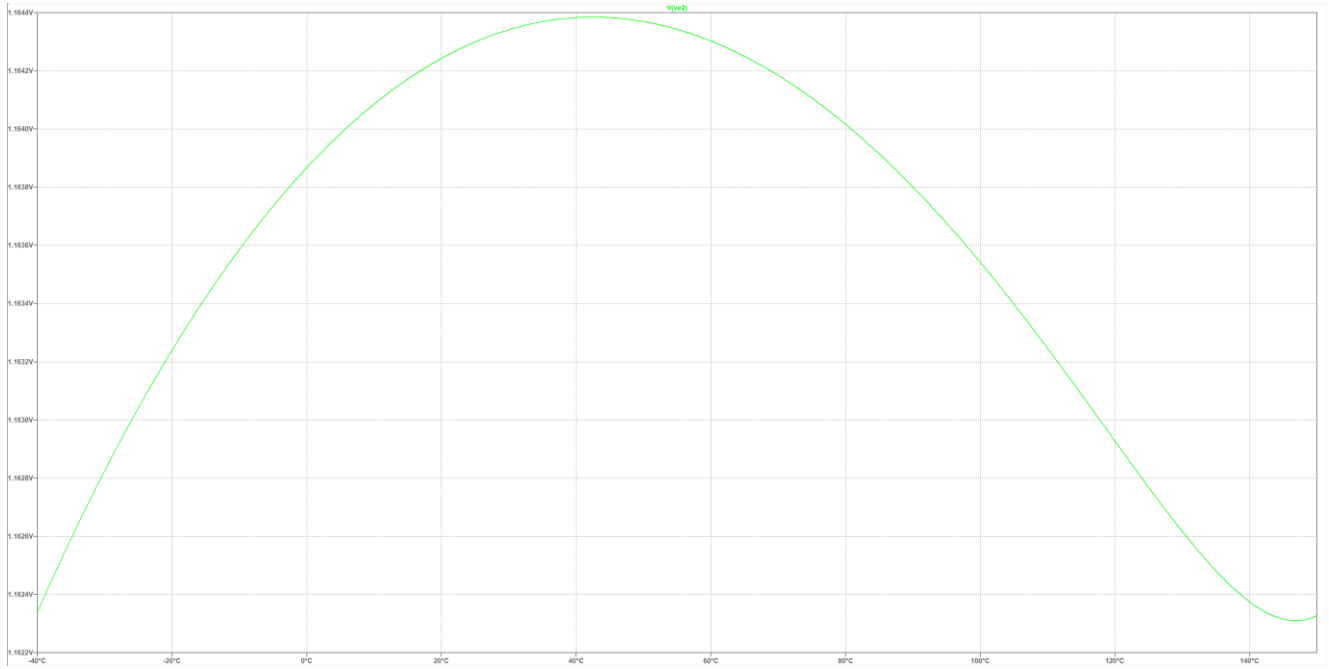


Figura 7. Rezultat analiză de temperatură

Variația cu temperatura: $\frac{\partial V_{BG}}{\partial \theta}$=2mV

2.2. Bandgap vs. PVT

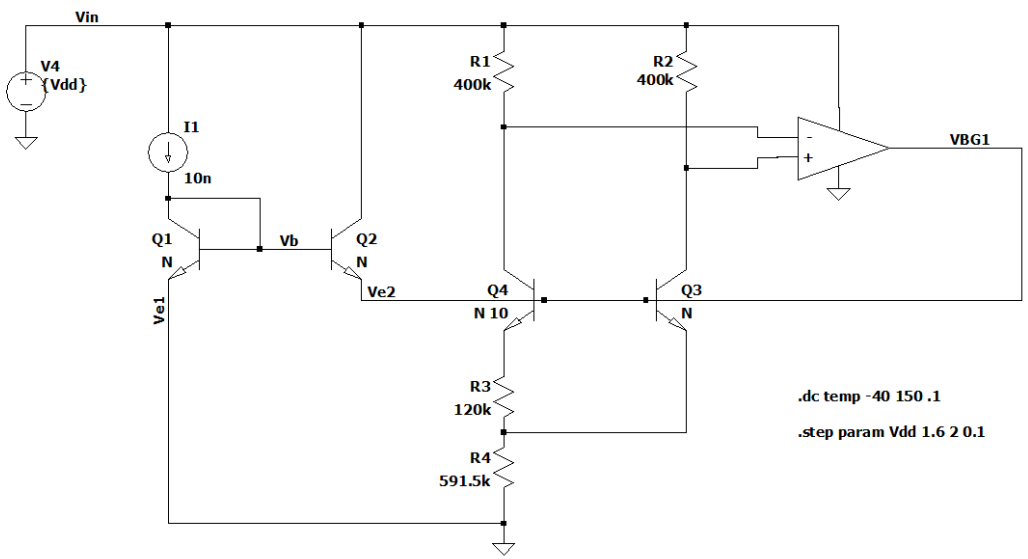


Figura 8. Circuit Bandgap + analiză de temperatură + variația alimentării

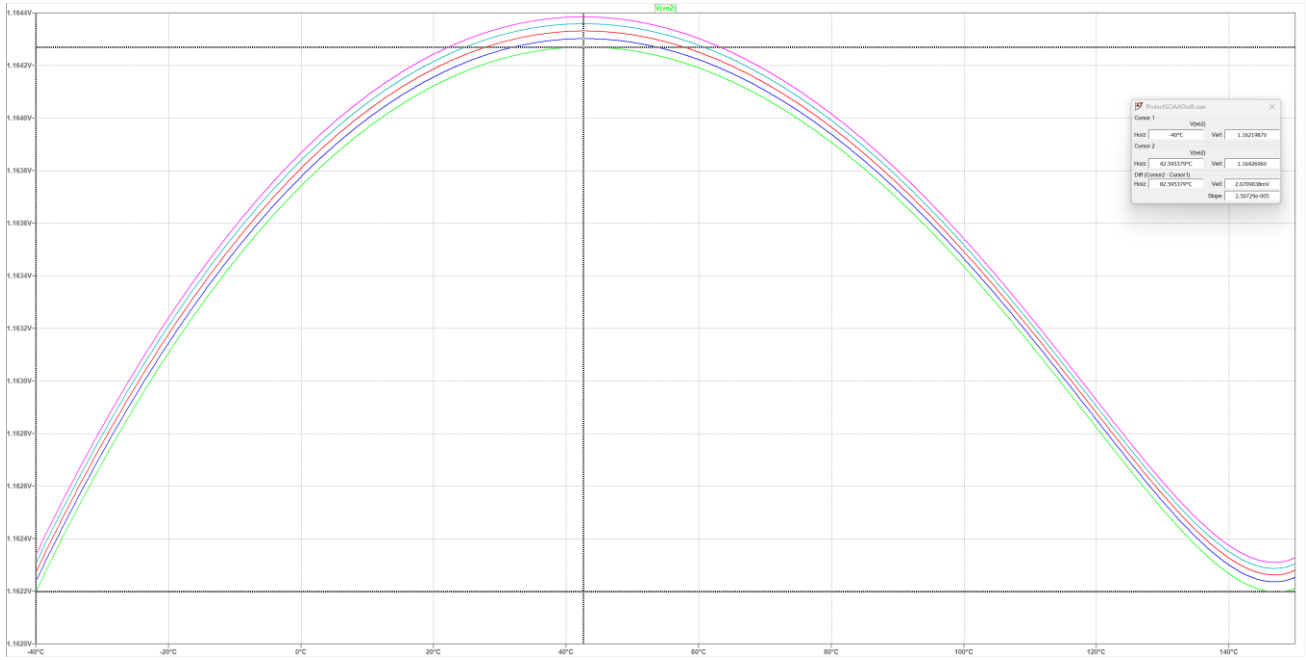


Figura 9. Rezultat analiză de temperatură + variația alimentării

Variația cu temperatura: $\frac{\partial V_{BG}}{\partial \theta}$=2mV $\pm 3\% \times 1.2V = 36mV$

Variația cu sursa de alimentare: 33µV

Variația cu toleranța rezistențelor (1%): 22mV

2.3.1 Analiza Monte Carlo la -40°C

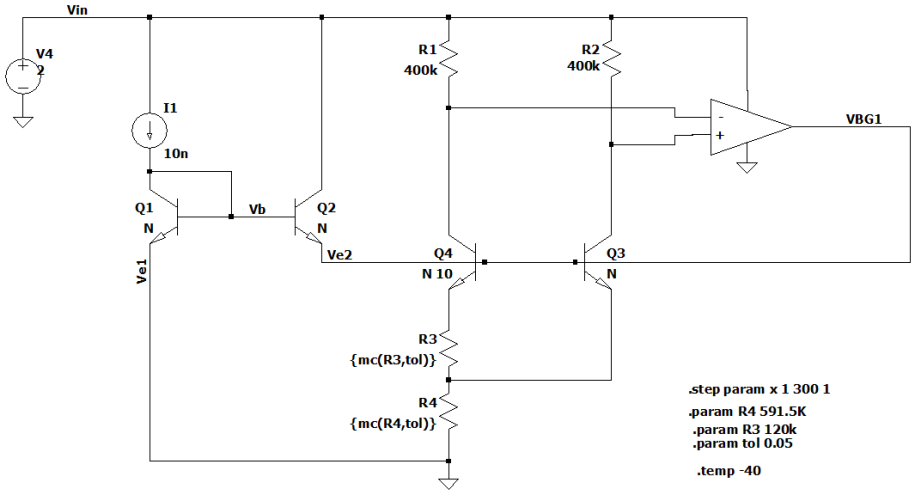


Figura 10. Circuit Bandgap + analiză Monte Carlo la -40°C

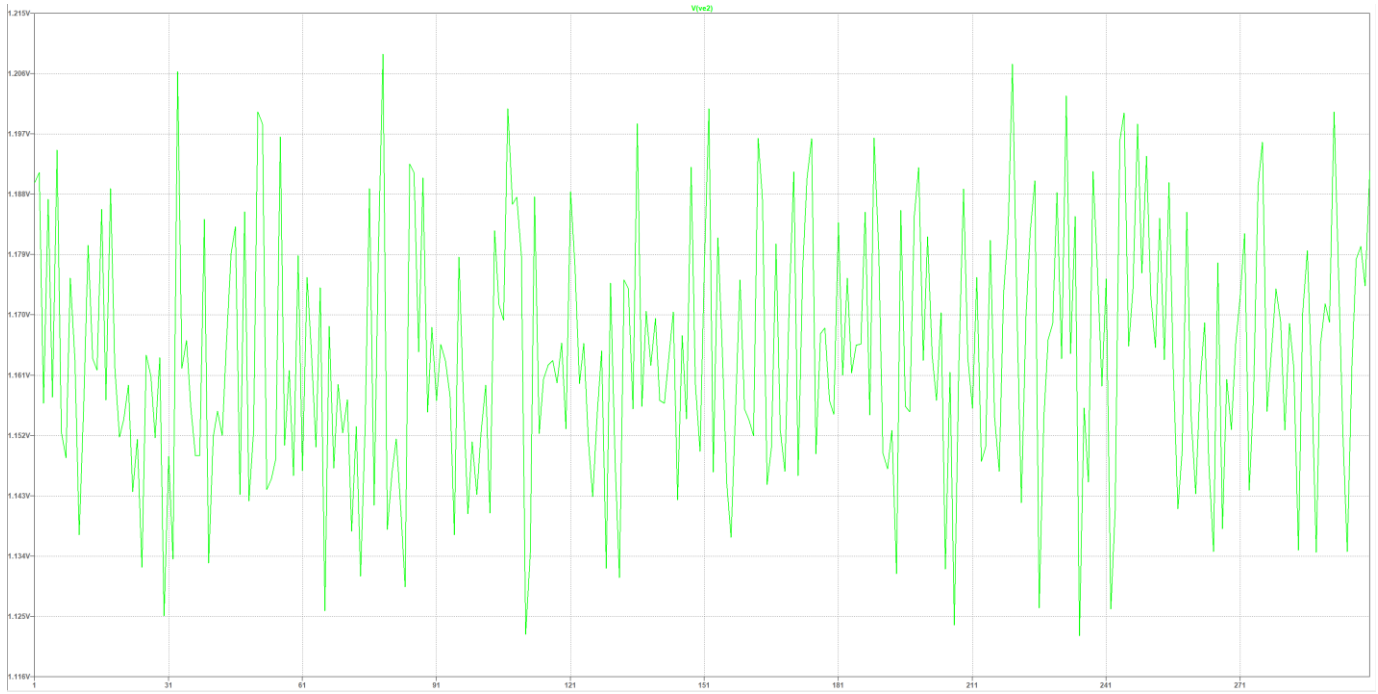


Figura 11. Rezultat analiză Monte Carlo la -40°C

2.3.1 Analiza Monte Carlo la 25°C

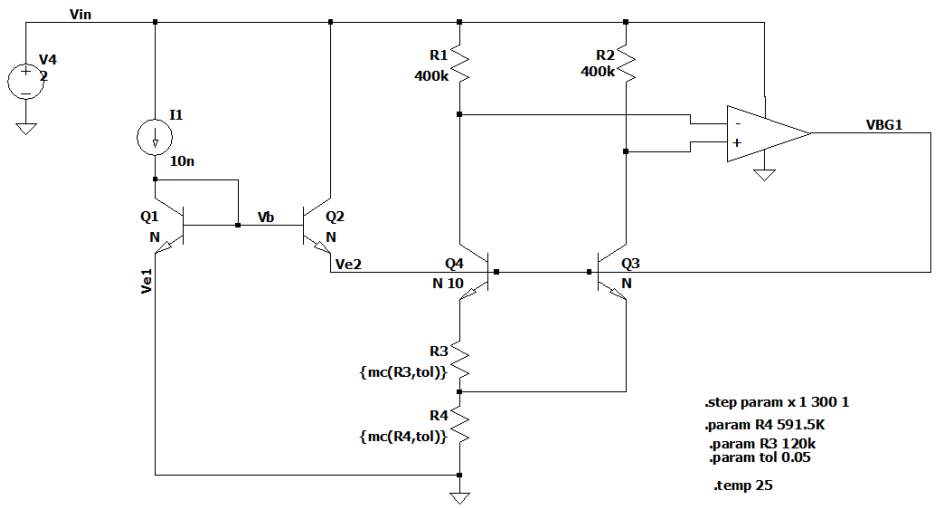


Figura 12. Circuit Bandgap + analiză Monte Carlo la 25°C

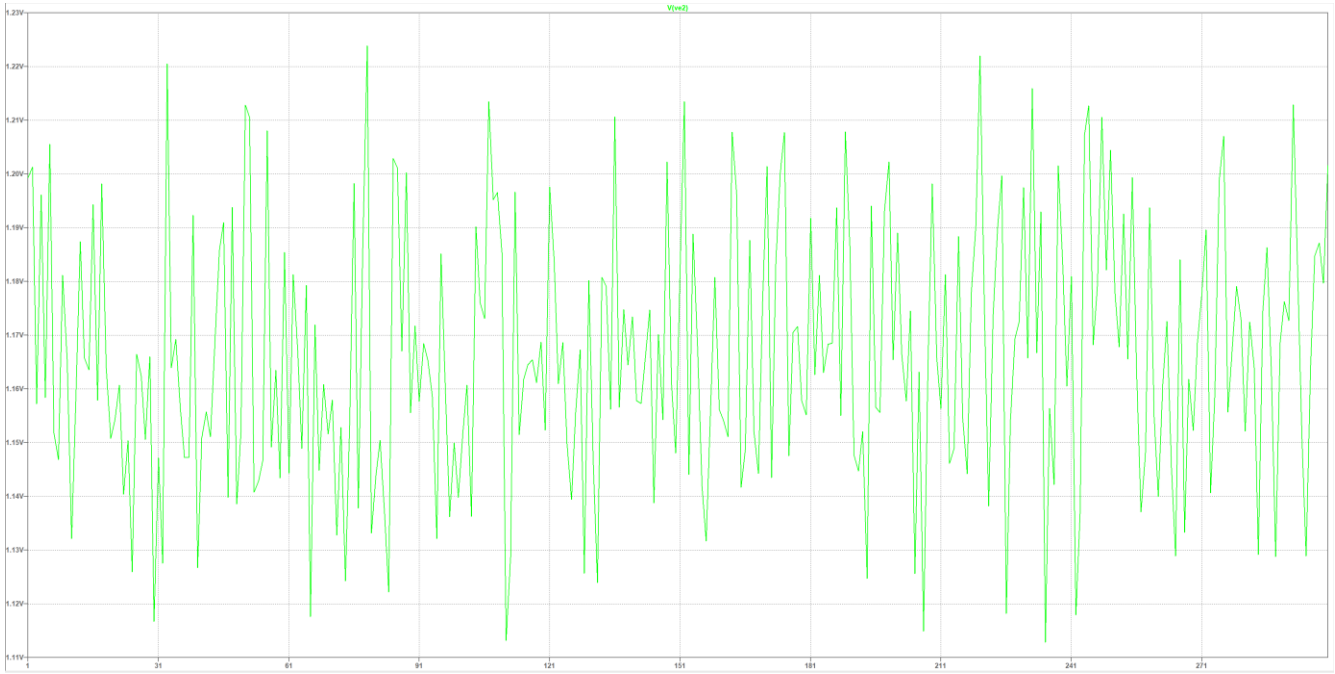


Figura 13. Rezultat analiză Monte Carlo la 25°C

2.3.1 Analiza Monte Carlo la 150℃

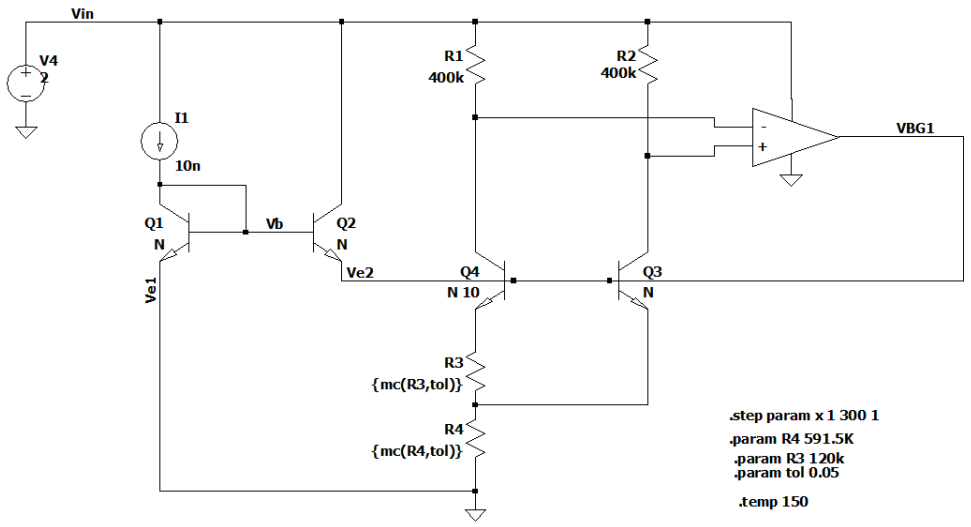


Figura 14. Circuit Bandgap + analiză Monte Carlo la 150℃

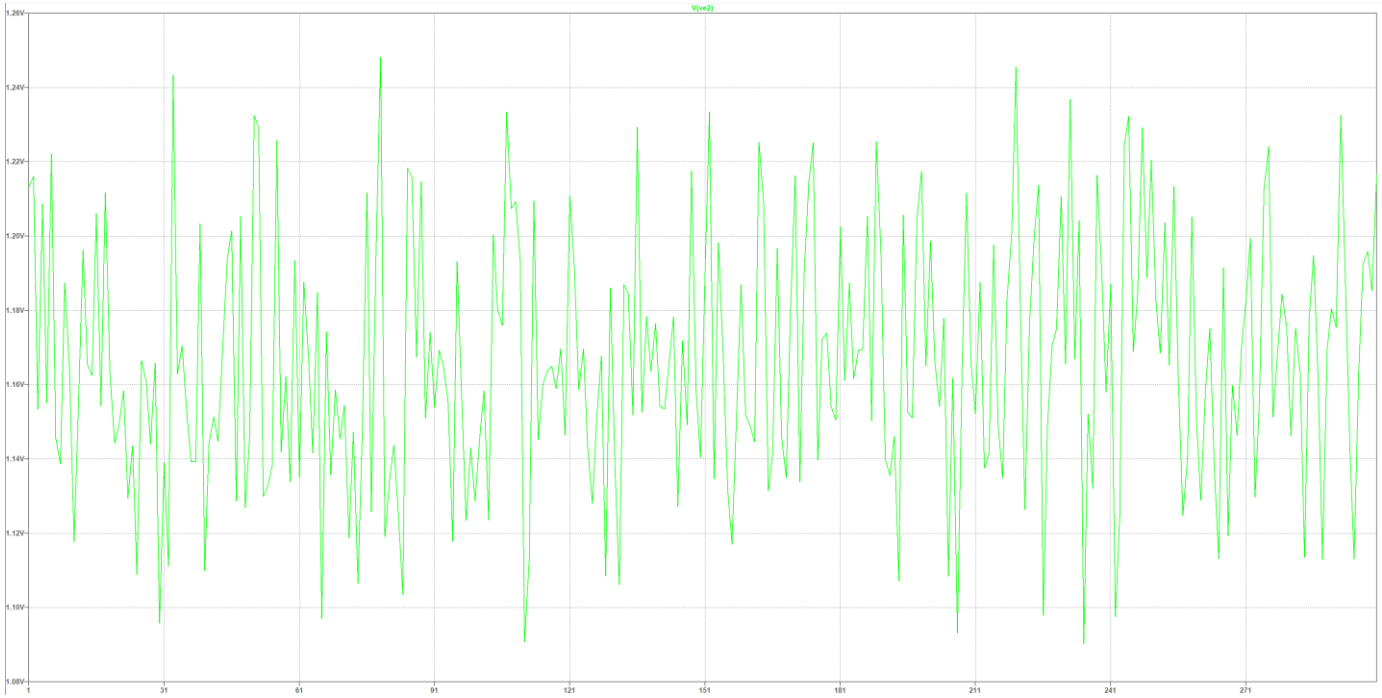


Figura 15. Rezultat analiză Monte Carlo la 150℃

2.4. Consum de curent

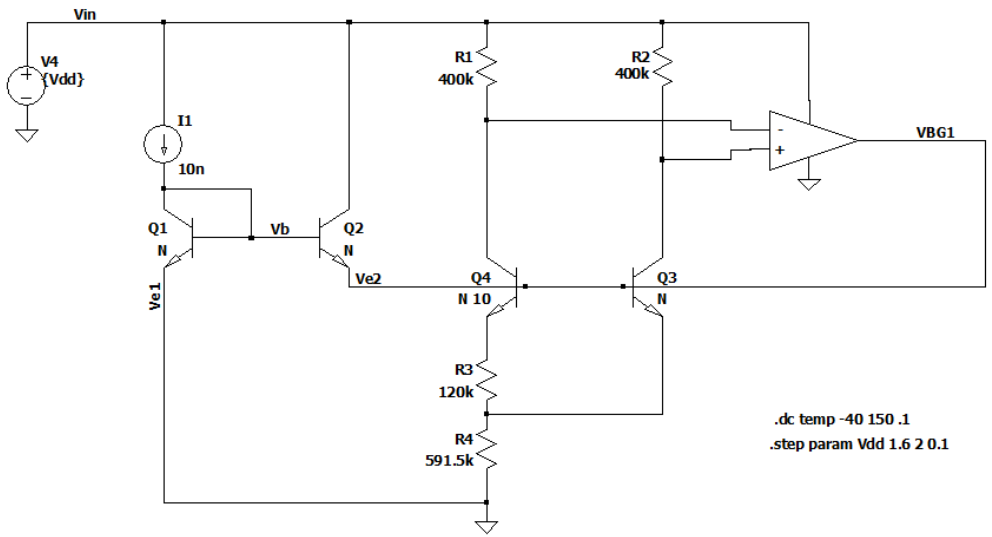


Figura 16. Circuit Bandgap + analiză de temperatură + variația alimentării

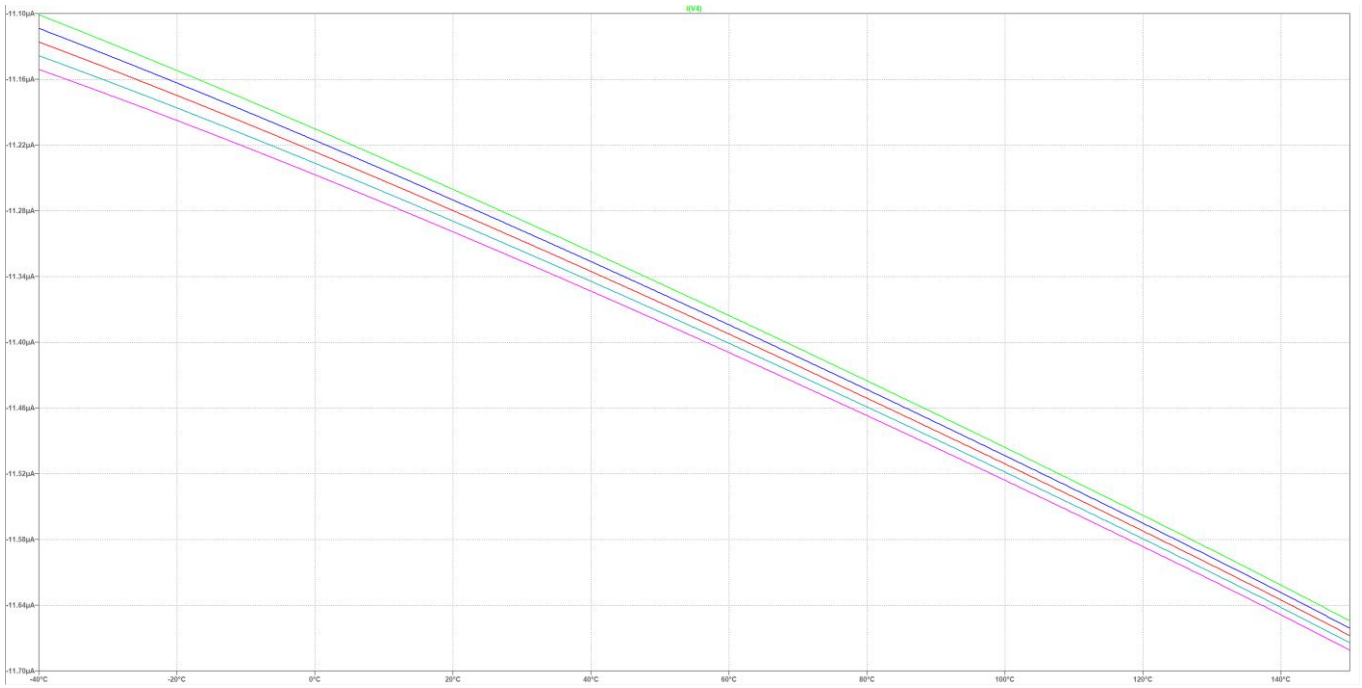


Figura 17. Curentul consumat

2.5. Stabilitate OTA vs. PVT

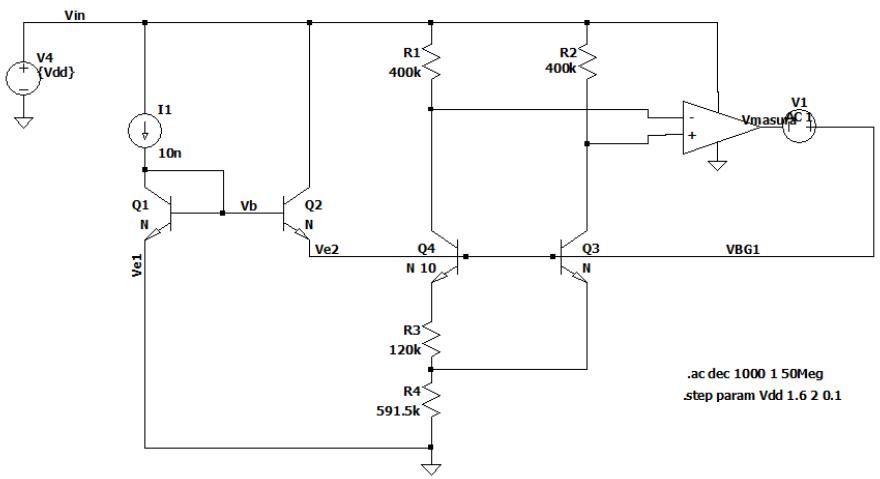


Figura 18. Circuit Bandgap + analiză ac

Figura 19. Amplificare la joasă frecvență

$$A0 \cong 69.25dB$$

2.5. Stabilitate OTA vs. PVT

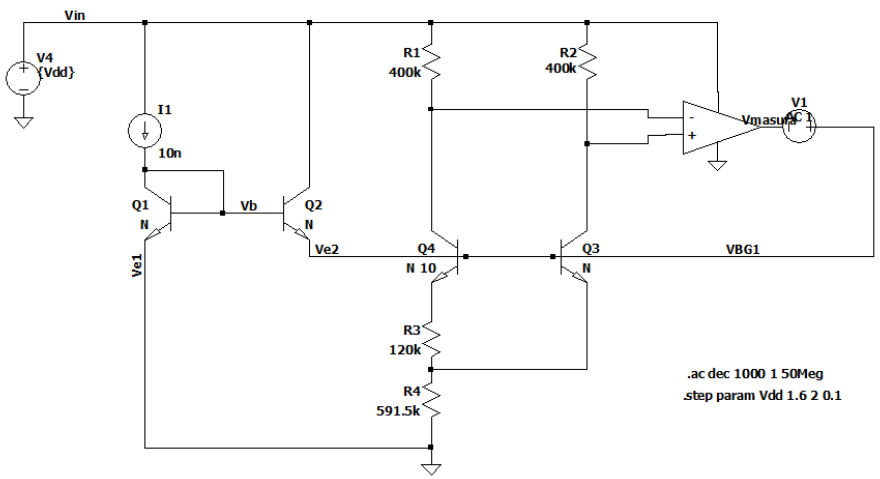


Figura 18. Circuit Bandgap + analiză ac

Figura 20. Faza măsurată la 0dB

PM=72.8° > 60°

2.5. Stabilitate OTA vs. PVT

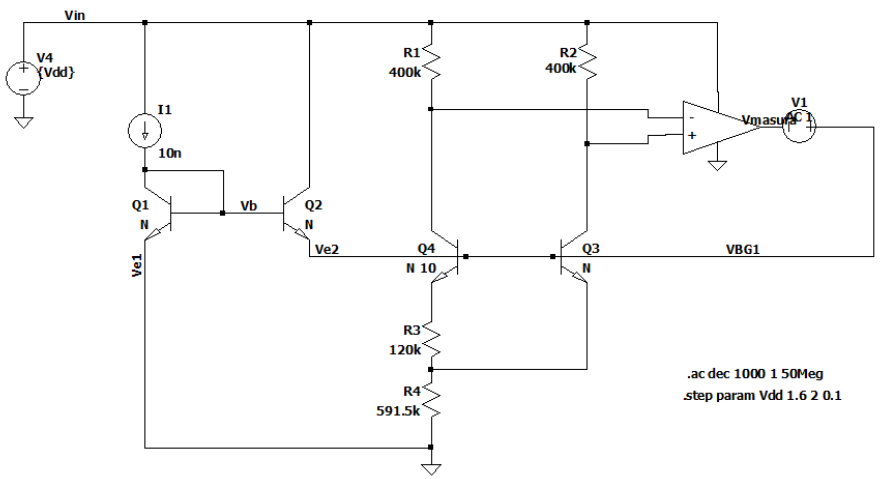


Figura 18. Circuit Bandgap + analiză ac

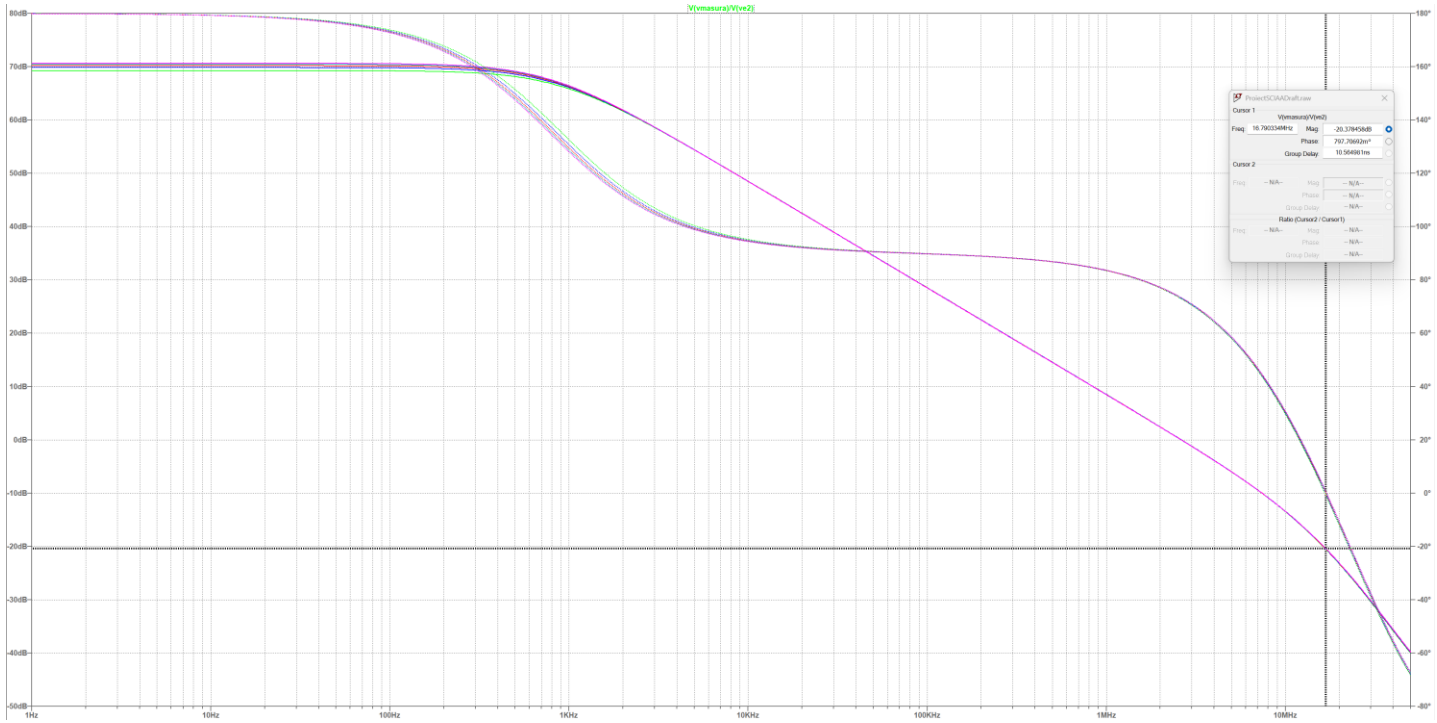


Figura 21. Câștigul măsurat la -180°

$$\mathrm{GM} = -20\mathrm{dB} < -10\mathrm{dB}$$

2.5. Stabilitate OTA vs. PVT

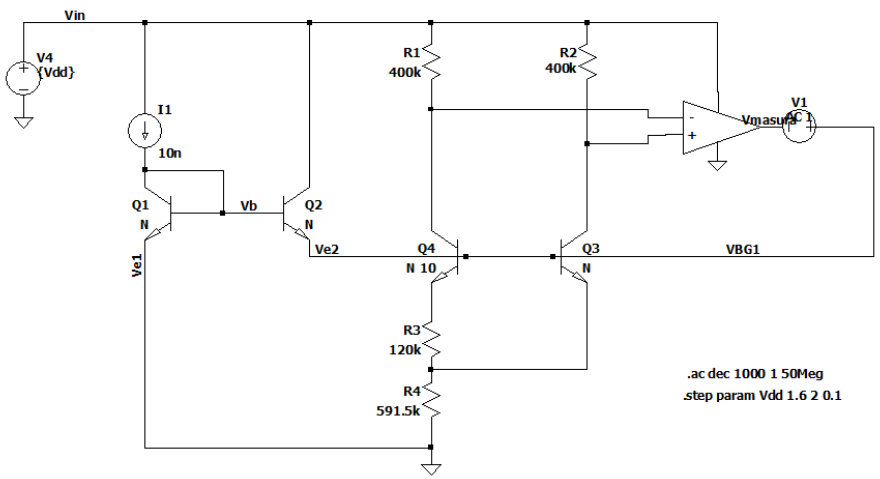


Figura 18. Circuit Bandgap + analiză ac

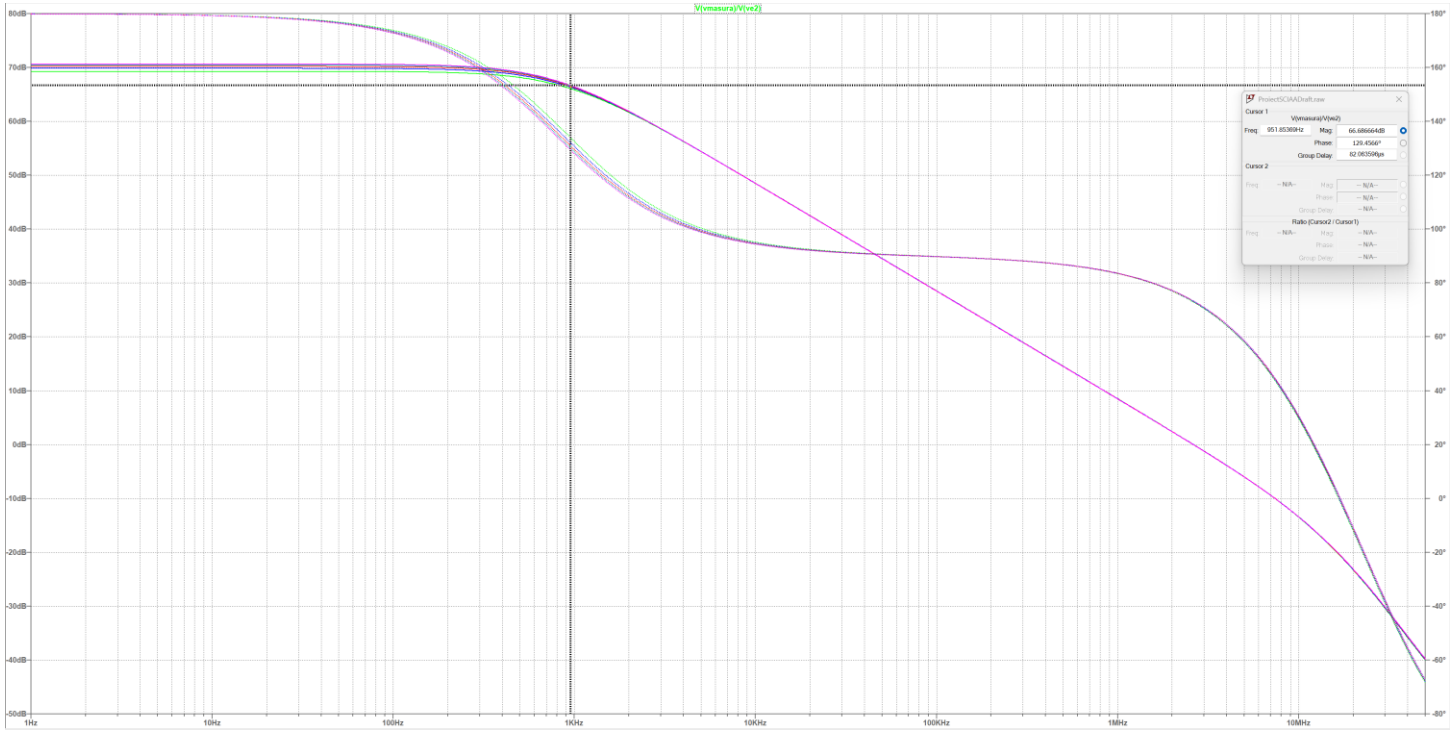


Figura 22. BW măsurat la −3dB

$$\mathrm{BW} \cong 950Hz$$

2.6. Timpului de pornire a bandgapului

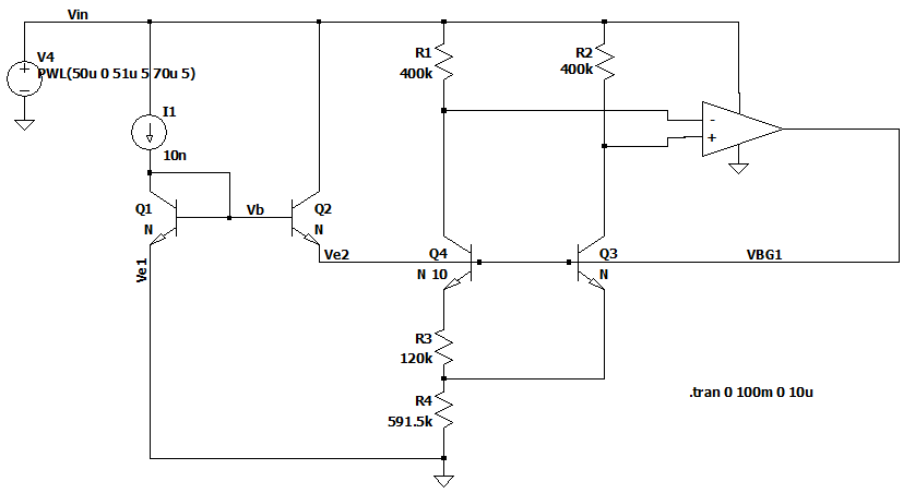


Figura 23. Circuit Bandgap + analiza timpului de pornire

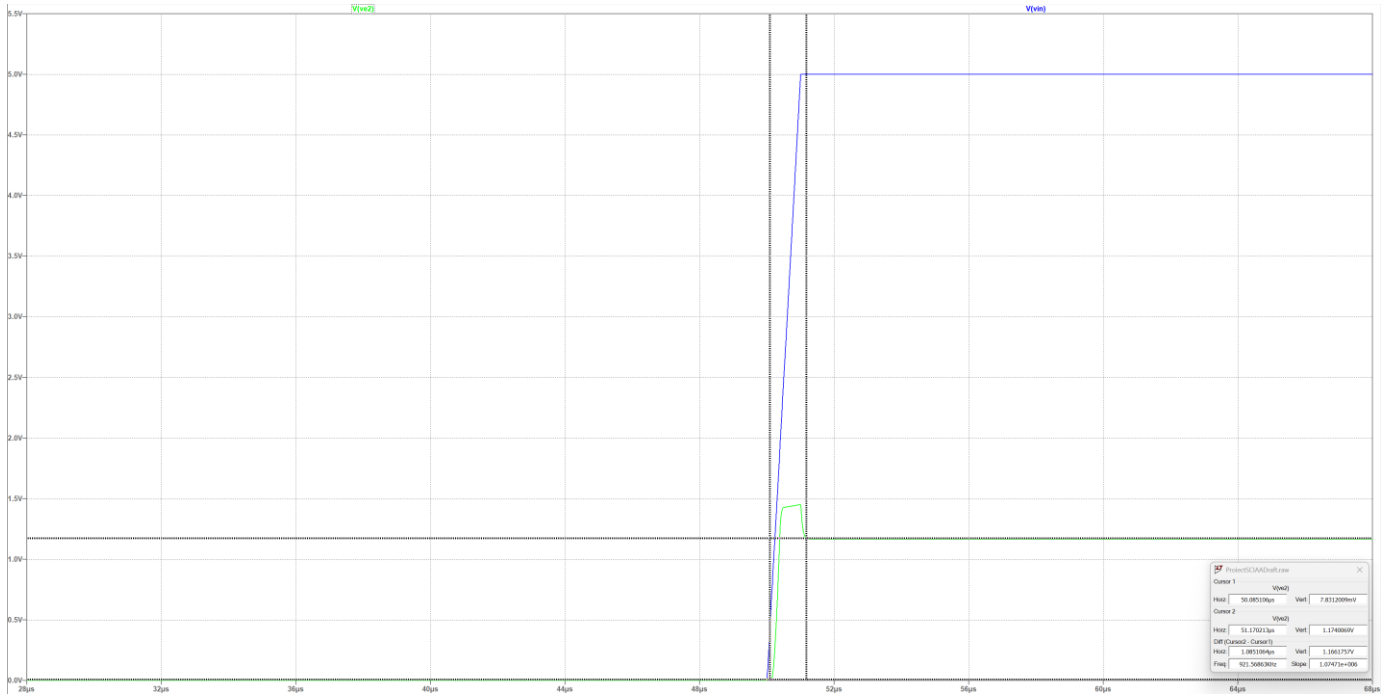


Figura 24. Rezultat timp de pornire

$$T_{rise} = 1\mu s \qquad t_{start-up} = 1.1\mu s$$

2.7. PSRR bandgap

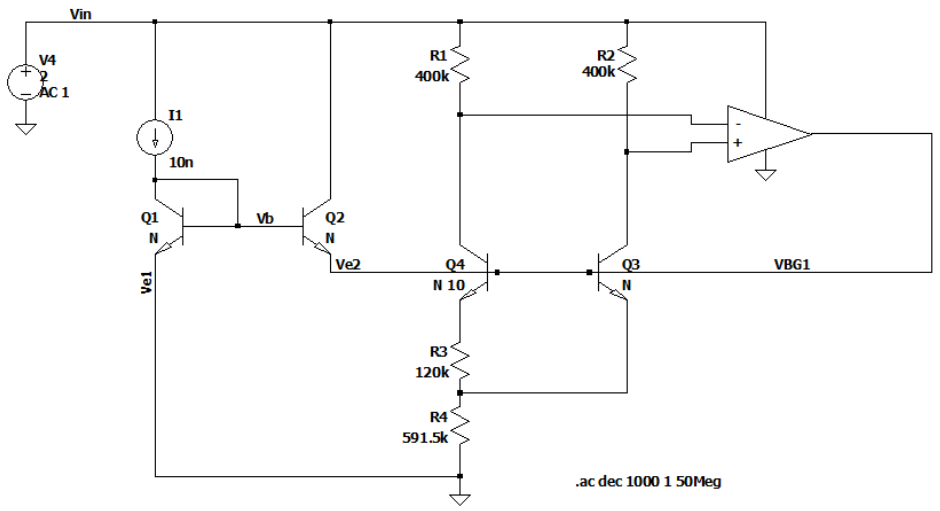


Figura 18. Circuit Bandgap + analiză ac

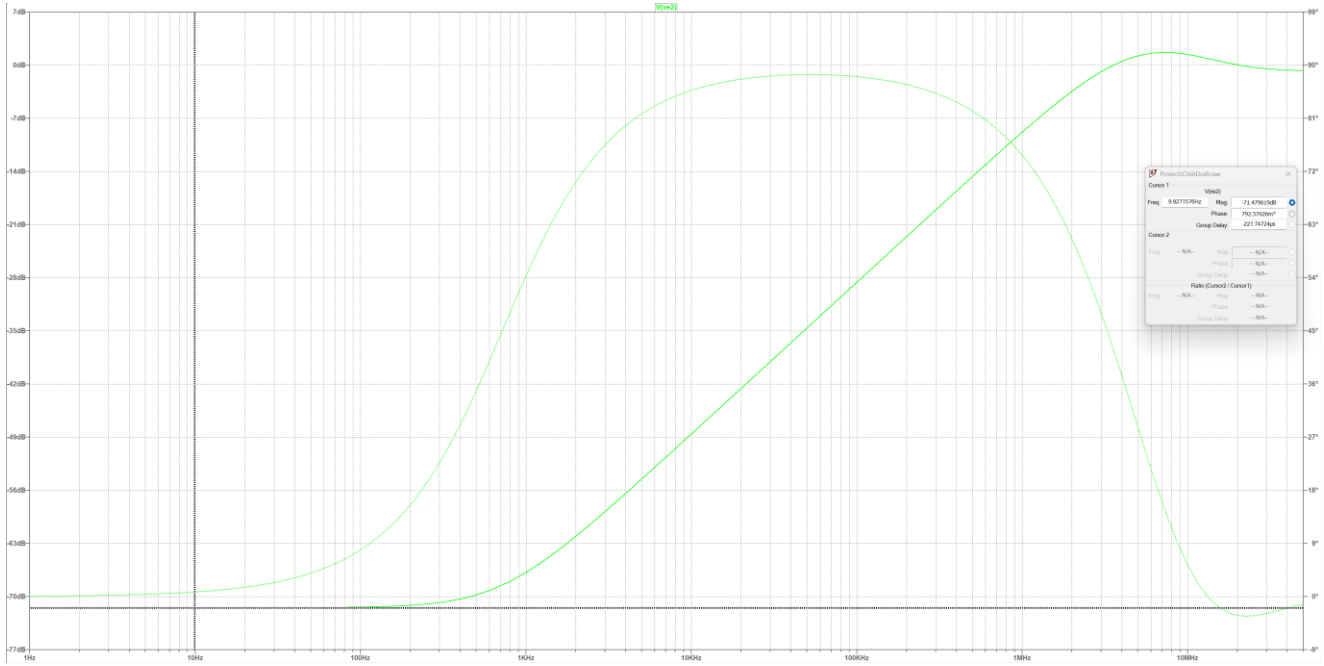


Figura 25. PSRR măsurat la joasă frecvență

PSRR=71.5dB @ 10Hz

3. Extra work

3.1. Compensare de ordinul 2

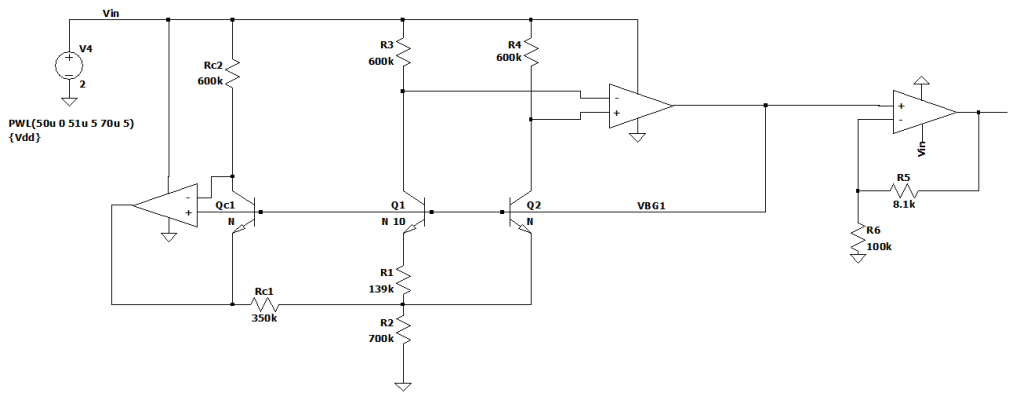


Figura 26. Circuit Bandgap compensare ordin 2

$$\frac{\partial V_{BG}}{\partial \theta} \cong constant \Rightarrow V_{B_{Qc1}} = constant$$

Amplificatorul A2 forteaza tensiunea colector-baza a transistorului Qc1 la valoarea zero. $V_{B_{Qc1}} = V_{C_{Qc1}}$

$$I_{C_{Qc1}} = \frac{V_{in} - V_{C_{Qc1}}}{R_{C2}} = constant \text{ (independent de temperatura)}$$

Factorul erorii de ordinul doi al tranzistorului Q2 este (XTI-1) iar al tranzistorului Qc1 este XTI.

$$V_{BE2} = V_{G0} - [V_{G0} - V_{BE}(T_0)]\frac{T}{T_0} - (XTI - 1)\frac{kT}{q}\ln\frac{T}{T_0}$$

$$\Delta V_{BE} = \frac{kT}{q}\ln N \ (nu\ introduce\ neliniaritate)$$

$$V_{BG} = V_{BE2} + a\Delta V_{BE} \Rightarrow -(XTI - 1)\frac{kT}{q}\ln\frac{T}{T_0} = termen\ neliniar$$

Principiu de funționare:

Adăugarea unui termen neliniar de semn opus pentru compensare

3.1. Compensare de ordinul 2

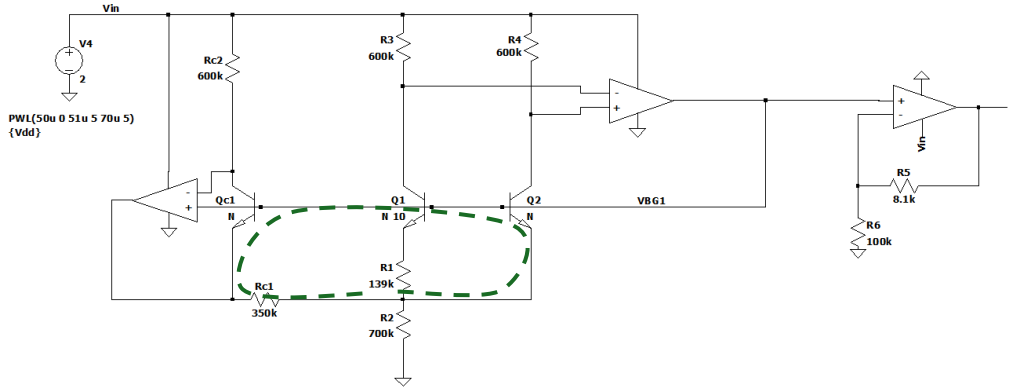


Figura 26. Circuit Bandgap compensare ordin 2

$$V_{BG} = V_{BE2} + V_{R3}$$

$$V_{R3} = R_2(I_{PTAT} + I_{NL})$$

$$I_{NL} = I_{Rc1} = \frac{V_{BE}{}^{PTAT} - V_{BE}{}^{ZTC}}{R_{C1}}$$

$$V_{BE}{}^{PTAT} - V_{BE}{}^{ZTC} = V_{G0} - [V_{G0} - V_{BE}(T_0)]\frac{T}{T_0} - (XTI - 1)\frac{kT}{q}\ln\frac{T}{T_0} - \\ - \left(V_{G0} - [V_{G0} - V_{BE}(T_0)]\frac{T}{T_0} - XTI\frac{kT}{q}\ln\frac{T}{T_0}\right) \\ = \frac{kT}{q}\ln\frac{T}{T_0}$$

$$I_{NL} = \frac{1}{R_{C1}}\frac{kT}{q}\ln\frac{T}{T_0}$$

3.1. Compensare de ordinul 2

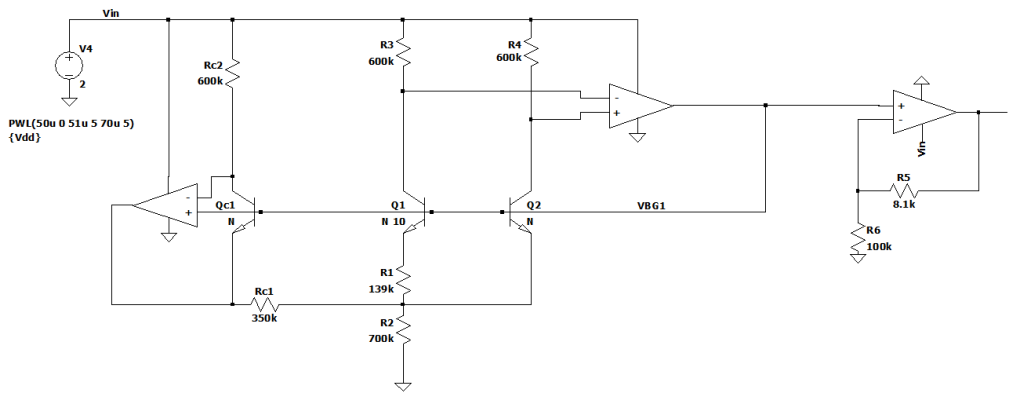


Figura 26. Circuit Bandgap compensare ordin 2

$$V_{BG}(neliniar) = -(XTI - 1)\frac{kT}{q}\ln\frac{T}{T_0} + \frac{R_2}{R_{C1}}\frac{kT}{q}\ln\frac{T}{T_0}$$

$$V_{BG}(neliniar) = 0 \Rightarrow \frac{R_2}{R_{C1}} = XTI^* - 1 = 2 \Rightarrow$$

$$\Rightarrow \begin{cases} R_{C1} = 350K\Omega \\ R_2 = 700K\Omega \end{cases}$$

* *LTspice uses default* $XTI = 3$

3.1. Compensare de ordinul 2

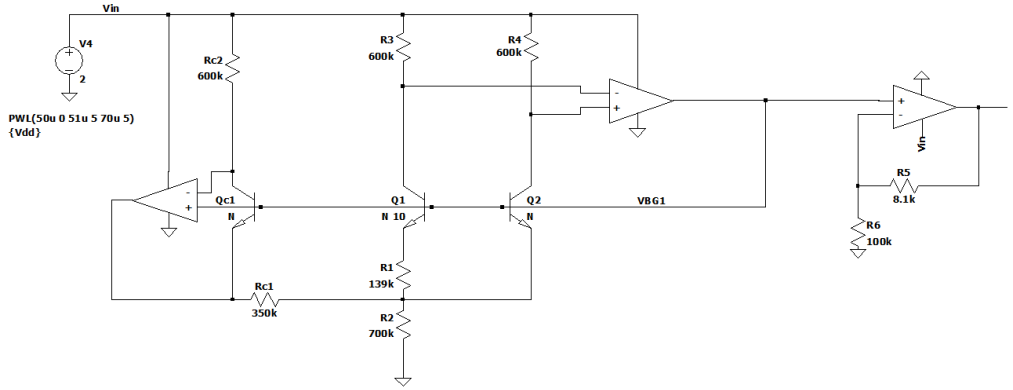


Figura 26. Circuit Bandgap compensare ordin 2

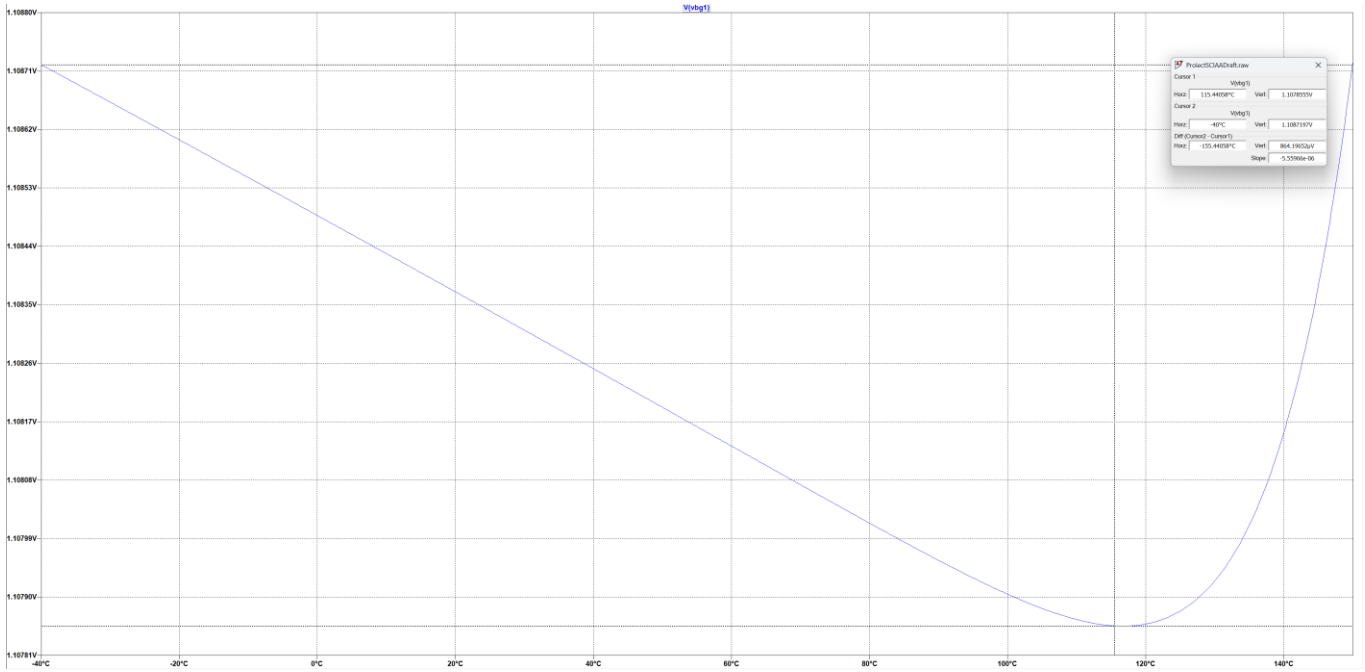
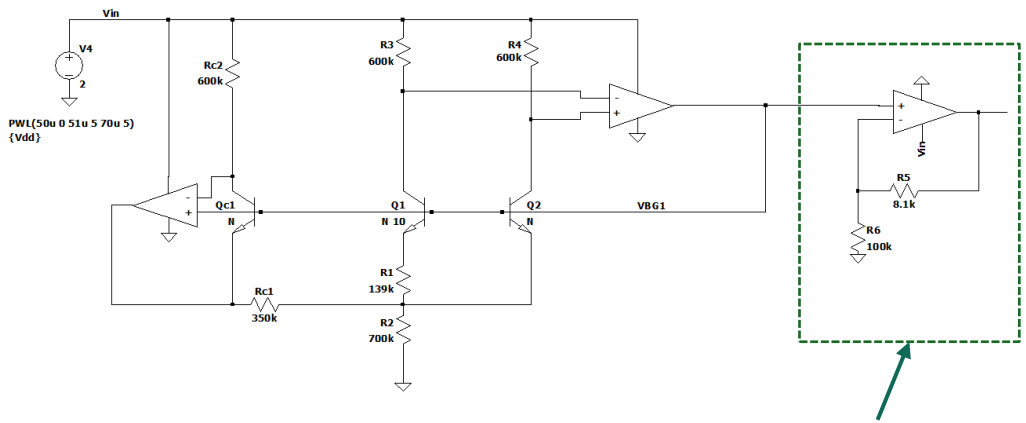


Figura 27. Rezultat analiză temperatură după compensare

Variația cu temperatura: $\frac{\partial V_{BG}}{\partial \theta}$=864μV

3.1. Compensare de ordinul 2



Etaj de amplificare pentru a obține 1.2V

Figura 28. Circuit Bandgap + Ridicarea nivelului de tensiune la ieșire cu etaj de amplificare

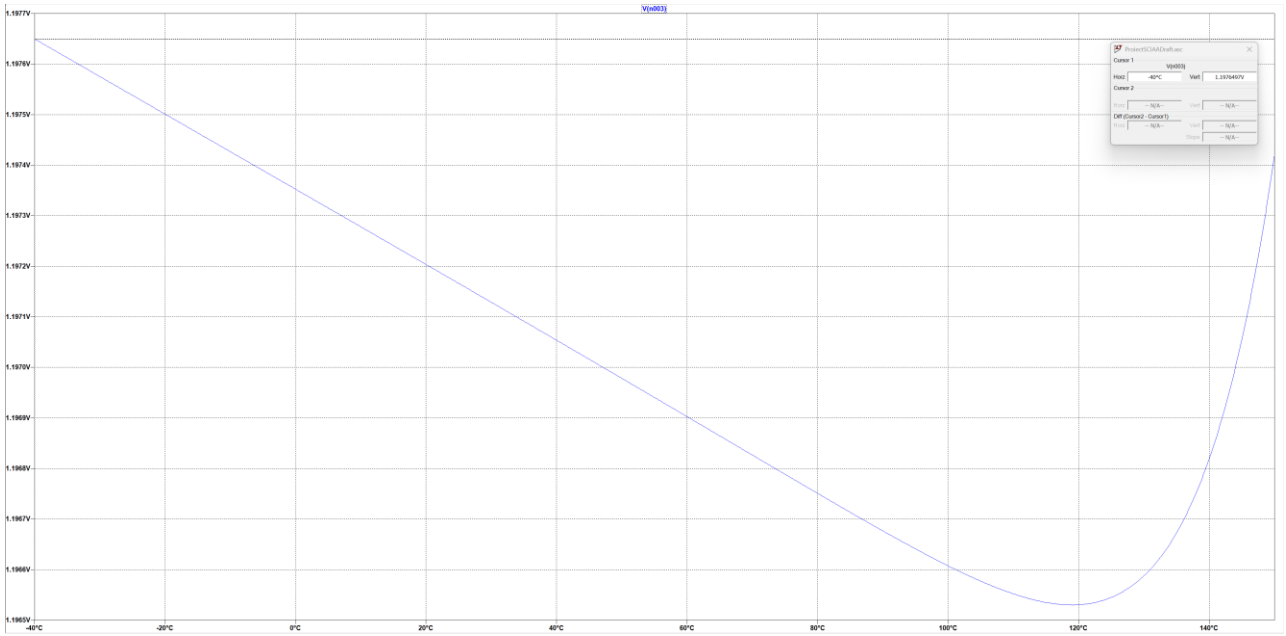


Figura 29. Rezultat final (varianta 1)

3.1. Compensare de ordinul 2

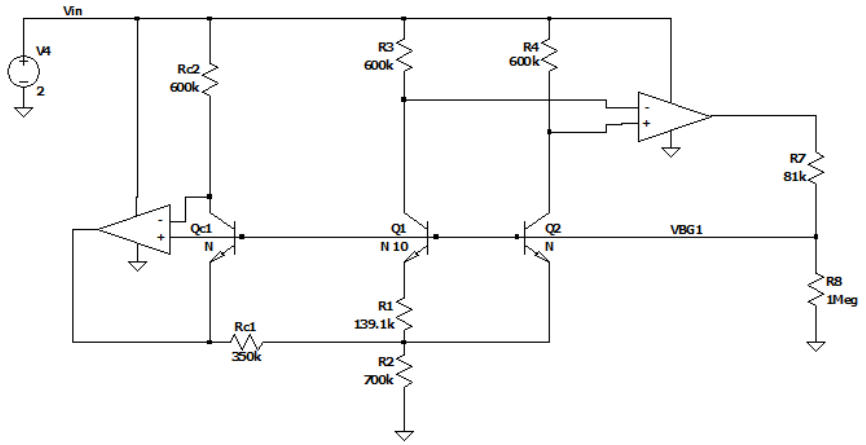


Figura 30. Circuit Bandgap + Ridicarea nivelului de tensiune la ieșire cu divizor rezistiv

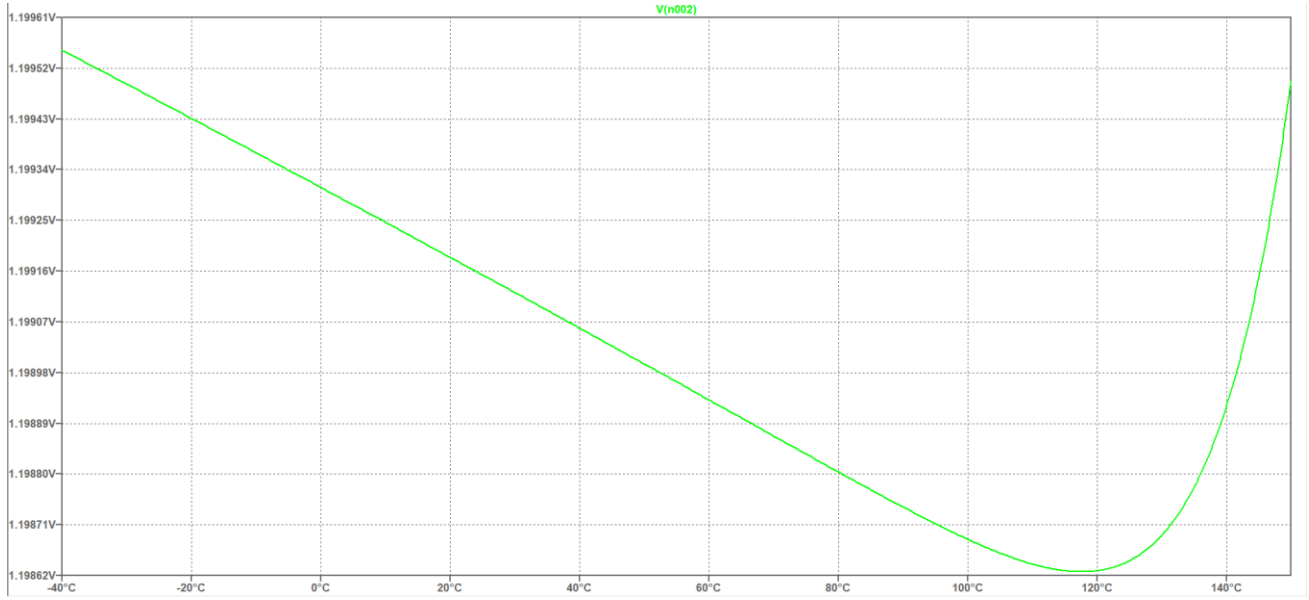


Figura 31. Rezultat final (varianta 2)

Mulțumesc pentru atenție!